

MICHAEL EMEFIENEM

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SOFTWARE ENGINEER

Software Engineer with 2+ years of experience building production-grade web applications using React, TypeScript, Node.js, and PostgreSQL. Experienced in owning features end-to-end across public-sector platforms, startup products, and AI-powered applications. Proven ability to deliver scalable frontend systems, real-time experiences, and secure full-stack solutions that improve operational efficiency and user experience.

TECHNICAL SKILLS

- **Frontend:** React, Next.js, Vue.js, TypeScript, JavaScript, Tailwind CSS, ShadCN
- **Backend:** Node.js, Express.js, PostgreSQL, Prisma ORM, Drizzle ORM
- **State Management:** Zustand, Redux, React Query, RTK Query
- **Tools & Platforms:** Git, Docker, AWS, Postman, Cursor, Claude, AI-assisted development workflows, Vitest, Jest, React Testing Library

PROFESSIONAL EXPERIENCE

Axendit Limited – Software Engineer

Lagos, Nigeria | March 2024 – April 2026

School Management System (Multi-role SaaS Platform)

Frontend Engineer

- Engineered role-specific dashboards and UI flows for 6 user types using Next.js, TypeScript, Redux, and RTK Query for efficient server-state management.
- Implemented a real-time communication layer using WebSockets, enabling instant messaging and live announcements across all user groups.
- Built and maintained 10+ feature modules including attendance tracking, grade management, course enrollment, finance, e-library, transportation, and complaint management systems.
- Implemented JWT-based authentication and role-based access control (RBAC), securing routes and dynamically rendering features based on user permissions.
- Collaborated with another frontend developer to deliver UI components, API integration, and role-based workflows.

Document Management System

Software Engineer (Backend Focus)

- Designed and developed RESTful APIs using Node.js, Express.js, PostgreSQL, and Prisma.
- Built document storage, folder hierarchy, version control, and fine-grained access control for 3 user roles.
- Implemented Redis caching to improve document retrieval and search performance.
- Developed an audit trail system tracking uploads, shares, permission changes, and document lifecycle events.

NCAA Consumer Protection Portal – Version 2

Frontend Engineer / Contributor

- Co-developed Version 2 of a nationwide consumer complaint platform for the Nigeria Civil Aviation Authority.
- Improved frontend workflows for complaint submission, tracking, and resolution.
- Collaborated with cross-functional teams to translate requirements into production-ready features.

Engineering Practices & Collaboration

- Wrote unit and integration tests using Vitest to improve system reliability and edge-case coverage.
- Participated in architecture discussions and code reviews to improve scalability and maintainability.
- Collaborated with product and backend teams to deliver end-to-end features aligned with business requirements.

Bond – Frontend Developer (Contract)

Remote | January 2026 – March 2026

- Led frontend development for the company's marketing website and internal back-office platform using React and TypeScript.
- Designed reusable component systems and scalable frontend architecture that accelerated feature development.
- Built merchant dashboard functionality that improved transaction visibility and streamlined operational processes.
- Integrated frontend applications with backend APIs while maintaining clean state management and modular code organization.
- Delivered production-ready features under tight deadlines in a fast-paced startup environment.

Zidio Development – Web Developer Intern

Remote | June 2023 – February 2024

- Developed responsive web applications using React, JavaScript, and Redux, improving user experience across desktop and mobile devices.
- Collaborated with senior developers to deliver client-facing functionality and resolve production issues.

SELECTED PROJECTS

AI Workflow Automation Platform | Next.js, TypeScript, React Flow, tRPC, Prisma, Postgres, Inngest, Better Auth

- Built an AI-powered no-code workflow automation platform that enables users to visually design and execute complex business automations through a drag-and-drop interface inspired by Zapier and n8n.
- Designed a node-based workflow execution engine supporting webhook, scheduled, and manual triggers, with extensible integrations for external services and APIs.
- Integrated third-party AI APIs including OpenAI, Claude, and Gemini, building resilient abstractions to support content generation, summarization, and classification workflows.
- Implemented reliable background job processing using Inngest to execute asynchronous workflows and long-running tasks at scale.
- Developed subscription and usage control features, including authentication, access management, and payment integration for SaaS monetization.
- Architected the application using end-to-end TypeScript with tRPC, Prisma, and Neon Postgres to ensure type safety, maintainability, and scalability.

AI Wireframe-to-Code Platform | Next.js, TypeScript, OpenRouter AI, PostgreSQL, Drizzle ORM, Tailwind CSS

- Built an AI-powered wireframe-to-code platform that converts uploaded UI wireframes into editable React and Tailwind CSS components.
- Integrated OpenRouter AI models to generate frontend code dynamically from uploaded wireframe inputs.
- Developed an interactive browser-based editing workflow using CodeSandbox integration.
- Implemented TypeScript-based scalable architecture and PostgreSQL data persistence with Drizzle ORM.

Live Demo: <https://wireframe-to-code-emefienem.vercel.app/>

AI Waste Management System | Next.js, Google Gemini AI, PostgreSQL and Drizzle ORM

- Built an AI-assisted waste reporting platform that uses Google Gemini AI to verify uploaded waste reports.
- Developed reward and leaderboard systems to improve community participation and engagement.
- Implemented secure authentication flows with Web3Auth and optimized data handling using Drizzle ORM.
- Built responsive user interfaces with Tailwind CSS and React Hooks for improved usability across devices.

Live Demo: <https://ai-waste-management-system.vercel.app/>

EDUCATION

LAGOS STATE UNIVERSITY

Lagos, Nigeria | July 2021 – September 2025

Bachelor of Science (B.Sc.) Computer Science – CGPA: 4.36/5.0

